

Hane Lee (they/them)

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(+1) 857-500-3241

ACADEMIC POSITIONS

University of Chicago Data Science Institute

Postdoctoral Research Scientist, Data and Democracy Lab

Chicago, IL

2025 – Present

EDUCATION

Columbia University

PhD in Statistics

New York, NY

2019 – 2025

Committee: Michael Sobel (chair), Naoki Egami, Robert Shapiro, and Zhiliang Ying

Massachusetts Institute of Technology

MS in Media Arts and Sciences, MIT Media Lab, Opera of the Future

Cambridge, MA

2017 – 2019

BS in Electrical Engineering, minor in Music

2013 – 2017

RESEARCH INTERESTS

Political methodology; American politics; public opinion and polarization; race, ethnicity, and politics.

PUBLICATIONS

Forthcoming in Annals of Applied Statistics. “[Measuring Public Opinion: “The Wasserstein Bipolarization Index”, with Application to Cross-National Attitudes Toward Mandatory Vaccination for COVID-19](#)”, with Michael Sobel.

Chris Andrade, Jonathan Auerbach, Icaro Bacelar, Hane Lee, Angela Tan, Mariana Vazquez, and Owen Ward (2023). “Does it pay to park in front of a fire hydrant?” *Significance* 20(1), pp. 28–30.

WORKING PAPERS

Working paper. “The Polarizing Middle Ground: Independents in Public Opinion Polarization”.

Working paper. “Quantifying Group-Based Electoral Competitiveness”, with Yuki Atsusaka and Diana Da In Lee.

Working paper. “[Measuring Social Ties from Roll Call Votes: A Fused Latent Factor and Social Network Approach](#)”, with Andrew Davison and Zhiliang Ying.

RESEARCH & WORK EXPERIENCE

Research Assistant, Prof. Tod Machover, MIT Media Lab	2017-2019
- Designed and prototyped interactive augmented reality (AR) musical experiences - Assisted production of hybrid electronic and acoustic live performances and post-production editing	
Undergraduate Research Assistant, Prof. George Verghese, MIT RLE	2016-2017
- Developed simulation-based markers of drug titration levels to guide physicians during procedural sedation using spatial pharmacokinetic/pharmacodynamic models	
Intern, Ion Beam Applications (IBA) & Université Catholique de Louvain	2015
- Improved the collision detection system a proton therapy cancer treatment device, using Kinect point clouds and C++ based libraries PCL and CGAL.	
Undergraduate Research Assistant, Dr. Stefanie Shattuck-Hufnagel, MIT RLE	2014

TEACHING EXPERIENCE

Columbia University

Instructor

Calculus-based Introduction to Statistics	Summer 2024
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Teaching Assistant

Graduate

Probability Theory	Fall 2020, 2021, 2022, 2024; Spring 2021
Statistical Inference	Fall 2022, 2023, 2024
Accelerated Probability Theory/Statistical Inference	Fall 2022, 2024
Statistical Machine Learning	Spring 2022
Linear Regression Models	Spring 2023
Bayesian Statistics	Summer 2023

Undergraduate

Introduction to Statistics / Statistical Reasoning	Spring 2020 / Fall 2019
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CONFERENCE PRESENTATIONS

2025 Society for Political Methodology Annual Conference, *poster presentation*
2024 Society for Political Methodology Annual Conference, *oral presentation and discussion*
2023 Society for Political Methodology Annual Conference, *poster presentation*
2022 Minghui Yu Memorial Conference, *oral presentation*

SERVICE

PhD Student Representative, Columbia Statistics Department	2021 – 2022
Minghui Yu Memorial Conference organizer	2022
Diversity, Equity, and Inclusion Committee, Columbia Statistics Department	2020 – 2023
Department Representative, Columbia GSAS Student Council	2020 – 2021

SKILLS

Programming: *proficient* Python, R; *practical* MATLAB, javascript, C++

Languages: *native* English, Korean